# err.Error() to HTTP response enumeration (PX-F18, §11.4.276(C) class enumeration), rework 3 (2026-11-13)

| Field | Value |
|---|---|
| Date | 2026-11-13 |
| Roadmap item | PX-F18 (CAT-PREEXIST-3) |
| Review base | main @ 9ed6577b (the base the NO-GO review of 2026-11-12 read) |
| Current base | main @ 713a4f380 (every "HEAD" below) |
| Scope | every non-test .go file under catalog-api (all packages, `main.go` included) plus the other Go modules of the repository that serve HTTP (OCU-CUDA-Sidecar; `scripts/test-infra/nas_clients` scanned: 0 hits) |
| Status | enumeration only; no source file changed. Replaces the doc the review rejected (it was never landed) |
| Verdict labels | PROVISIONAL, review owed. Nothing here is a review GO |

## Why this is a rework

The NO-GO review (docs/FOLLOW_UPS_NEXT_PHASE.md, "PX-F18 enumeration doc review: NO-GO") found one unlisted class and five defects of the instrument and the wording. Each is closed in this document as follows (evidence is in the sections named).

| Review finding | Closure | Where |
|---|---|---|
| F1: `utils.SendErrorResponse(c,status,msg,err)` puts `err.Error()` into `details`; 109 call sites in 11 files (media_entity 35, collection 15, comic_pages 14, stats 10, media 10, search 6, recommendation 6, challenge 5, browse 5, media_browse 2, middleware/auth.go 1; auth.go:167 sends JWT parse text on 401) | The scanner now matches a bare `err` argument. At the review base it reproduces the review's numbers exactly: 109 bare-err call sites in the same 11 files with the same per-file counts, plus one more SendErrorResponse site that passes `err.Error()` as the *message* (collection_handler.go:335). The class is enumerated per site (rows B1..B226) and the mechanism is verified: base `utils/response.go:31` has `response.Details = err.Error()`; HEAD has no assignment to `Details` anywhere in `utils/` | "Class 1" and the per-site table |
| F2: internal/metrics/health.go:157 (unauthenticated /health/deep leaked the DB ping error); monitoring.go:251 (unwired) | Both are rows H1, H2. They are struct/map-literal flows that a sink-call scanner cannot see, so a second instrument (every zero-argument `.Error()` call, any context) was added. health.go:157 is closed at HEAD (fixed text, error logged). monitoring.go:251 is still present at HEAD and is unwired (no non-test importer of `catalogizer/internal/monitoring`) | "Class 2", rows H1/H2, "HEAD residual" R1 |
| F3: auth rows should be LEAK | Reclassified. At the review base `AuthService.Login` returns `fmt.Errorf("failed to get user: %w", err)` and similar chains (auth_service.go:105,147,170,177,183,189) and the handler sends `err.Error()` on 401, so the rows are LEAK, with authored sentinels (locked, disabled, pending) mixed in. The 20 former UNKNOWN rows were each resolved by reading the producer (rule list in "Classification rules") | per-site table, column Class |
| F4: middleware rows are latent | Reclassified LATENT (rows with `internal/middleware/middleware.go`). ErrorHandler is wired (`main.go:1038` at base, `:1045` at HEAD), but its two leaking branches need gin `ErrorTypeBind` or `ErrorTypePublic` errors, and no non-test code produces either (grep for `.Bind(`, `BindJSON`, `MustBindWith`, `AbortWithError(`, `ErrorTypePublic`: 0 hits outside the middleware file; `ShouldBind*` does not append to `c.Errors`). Third-party middleware that might append such errors was not enumerated: UNCONFIRMED | per-site table |
| F5: limits text wrong | The old text said "no such helper found by name search"; SendErrorResponse is exactly such a helper and 109 sites used it. "Limits of the instruments" is rewritten from what was measured | last section |

| Review finding | Closure | Where |
|---|---|---|
| F6: tool/main.go errSite misses bare `err` arguments | Fixed in instrument 1 (kind `BARE`). The control needle below fails on the old logic (bare err not reported) and passes on the new | "Control needles" |

### Round-2 review findings (NO-GO on rework 2), closure in this rework

| Finding | Closure | Where |
|---|---|---|
| N1: "class complete / no unclosed LEAK" false: class excluded state persisted and later served; sink scan missed an err inside a helper in a sink (fixture f) and a non-err-named variable in `%v` (fixture g) | Class redefined as error-derived text that can reach a client by ANY path. Instrument 1 gained kind `H:` (any call with an err-like identifier argument inside a sink argument) and err-like detection by type/nil-comparison; reviewer fixtures f and g are now found. New instrument 3 (flow scan) lists error-derived text stored in struct fields, map/kv entries, channels and call arguments. New section "Class 4" traces each stored field to its writer, route, gate and scrub. The "no LEAK row remains unclosed" sentence is withdrawn | Class definition, Instruments, Class 4, Control needles |
| N1 specific rows (GetStorageRootStatus, ListStoragePath, SyncSession.ErrorMessage, ScanStatus.Reason + event bus, scrubSecrets and sync_service.go:1057 census rows) | Rows S1 to S5 in Class 4; the census sentence "none reach a response" is corrected | Class 4, Census |
| N2: unrouted net/http variants counted as live | B7 to B10, B12 to B14, B80 to B83, B213, B214 reclassified LATENT with the routing evidence; totals recounted | Per-site table, Review-base totals |
| N3: validationText/paramRequiredText rows wrongly "CLOSED"; claim "0 SendErrorResponse sites carry error text" | HEAD status of VALIDATION rows reworded to "NOT CLOSED"; collection_handler.go:335,351, playlist_handler.go:95,188, service_handlers.go:258,298 and input_validation.go:403 added to R4; the claim is replaced | R4, Class 1 |
| N4: configuration_service.go:430,443, localization_service.go:849-904, recovery/retry.go:401, webdav_client.go:340-502 | Traced in Class 4 rows S6 to S10 | Class 4 |
| N5: B2 reason, B3 "(unresolved in review base)", ErrRecommendationsNotImplemented | B2 now VALIDATION with the producer cited; B3 reason cites auth_service.go:132; the sentinel text is noted in Class 1 | Per-site table, Class 1 |

# Instruments

Both are Go AST programs (go/parser, go/ast), built in a scratch directory; the sources are committed next to this doc as `docs/qa/err_error_class_scan/astscan_sinks.go.txt` and `all_error_calls.go.txt` (`.txt` so no Go tool picks them up). They were run on a clean clone at both bases, with each tree checked out without recursive submodule init.

1. **Rework 3 changes to the sink scan.** Err-like identifiers are now found by type (`error` in parameters, results and `var` declarations) or by comparison with nil, so a variable named `failure` or `cause` counts. Kind `H:` reports any non-`fmt` call that takes an err-like identifier inside a sink argument. Kind `F:` matches any err-like identifier argument of `fmt.Sprint*`.
2. **Sink scan.** For every non-test function it finds calls whose callee name looks like a response sink (names containing json, abortwith, string, write, send, respond, errorresponse, badrequest, servererror, encode, `http.Error`, html, data, emit, broadcast, notify, plus `c.Error`/`AbortWithError`); logger, `fmt.*`, `errors.*`, `strings.*` callees are excluded. An argument is reported when its subtree holds (E) a zero-argument `.Error()` call, (F) `fmt.Sprint*`/`Errorf` with an err-named identifier argument, (T) an identifier that the same function assigned from such an expression (taint, flow-insensitive), or (BARE) an err-named identifier passed as an argument.
3. **Flow scan** (`docs/qa/err_error_class_scan/flows_assign.go.txt`, skips `challenges/`). Lists error-derived text (`err.Error()`, `%v` of an err-like identifier, a variable assigned from either) assigned to a struct field, an index or map key, a key-value pair, a channel send or a call argument, with file, line, function and kind. Its rows are leads for Class 4, not verdicts.

4. **Census.** Every zero-argument `.Error()` call in every non-test file, with its syntactic context (argument of a named call, assignment, return, composite literal / key-value, branch, other). It catches flows instrument 1 cannot see (struct literals returned to a handler, helpers with unusual names).

### Control needles (§11.4.201(7)(b), §11.4.273)

- Known positive on the review base: `handlers/error_reporting_handler.go:85` (`http.Error(w, err.Error(), 500)`): found, 1 row, instrument 1. On HEAD that file has 0 rows (closed by CAT-PXF23-R5B).
- Synthetic fixture with four functions (bare err to `SendErrorResponse`; `fmt.Sprintf` into a variable then `c.JSON`; direct `err.Error()` in a map; `err.Error()` only in a log call and an == test): instrument 1 reported exactly the first three (kinds BARE, TEXT-via-taint, TEXT) and nothing for the fourth; instrument 2 reported the direct call (context kv), the log argument (`arg-of:log.Printf`) and the comparison (`branch`).
- Reviewer fixtures (round 2): `fixF` (`c.JSON(500, gin.H{"e": scrub(err)})`, helper nested in a sink) and `fixG` (`fmt.Sprintf("failed: %v", failure)` where `failure` is a non-err-named `error`) are both reported by the rebuilt instrument 1 (kinds `H:` and `F:`); good1 to good3 are still reported and bad1 is still silent. The previous build reported neither fixF nor fixG.
- Re-run of instrument 1 on a fresh checkout of each base reproduces the stored outputs: raw 432 rows (review base) and 371 rows (HEAD); 310 and 250 after removing the `strings.`, `sort.`, `bytes.` callee noise. New HEAD rows of kind `H:` are the helpers listed in Class 3 and R4 plus nil-comparison noise (`len(playlists)`, `[]string(p)` and similar, read and discarded as false positives).
- Fabricated token `ZZNEEDLE_NOPE`: 0 hits in all output files (instruments 1, 2 and 3, both bases).
- Cross-check of instrument 2 against `grep -rE '\.Error\(\)'` (non-test .go, catalog-api): review base AST 333 calls vs 329 grep lines; HEAD AST 255 vs 254. Line-set comparison (LC_ALL=C): AST-only lines 0 at both bases; grep-only lines are 1 at the review base and 4 at HEAD (universal_scanner.go:188 in both), and every one was read: they are comments that quote `err.Error()` (universal_scanner.go:188, auth_handler.go:36, client_errors.go:67, utils/response.go:31). The AST count exceeds the grep line count because several lines hold more than one call (5 such lines at each base).
- Taint is flow-insensitive: on HEAD `handlers/user_handler.go:122` is reported as `T:msg` because `msg` is assigned from `err.Error()` later in the function (line 168); at line 122 `msg` comes from `authClientMessage` (authored text). Read and confirmed a false positive; it is not counted below.

## Class definition

A site is in the class when text derived from an `error` value can reach a client (HTTP, WebSocket or SSE response body or header) by ANY path. Routes (rework 3 widens this; the earlier definition excluded server-side state that is persisted and later served): (1) `err.Error()` or `%v/%w/%s` of an error inside a response call; (2) a bare error passed to a helper that serialises it; (3) error text stored in a struct or map that a handler later serialises; (4) error text kept in a variable and used in a response; (5) error text returned from a helper into a response; (6) a helper called inside a sink argument (`gin.H{"e": scrub(err)}`); (7) `%v/%s` of an error-typed identifier without `.Error()`, whatever the identifier is called; (8) error text stored in a struct field, map, database row, session or status object and served later by another route (Class 4); (9) error text published on the event bus and forwarded to WebSocket or SSE clients. Sites that only log or branch on text are outside the class; sites that record text in server-side state are IN the class when a route serves it, and are traced in Class 4.

Classes used in the tables: **LEAK** (internal detail can reach the client: SQL/driver text, wrapped `%w` chains, paths, upstream or JWT parse text; "can include", not a demonstrated leak for every row: service and repository strings were read only where stated); **VALIDATION** (authored text that tells the caller which input to fix, or a sentinel's own text; may echo the caller's own input); **LATENT** (leaking code that no non-test caller can reach); **SAFE** (the err value does not reach the client: log-only, fixed-text helper, or a non-client sink); **CLOSED-BY-HELPER** and **CLOSED** are HEAD statuses, not classes.

### Classification rules (how the 20 old UNKNOWN rows were resolved)

Rule 1: a 4xx/5xx text produced by a service chain that contains `fmt.Errorf("...: %w", repoErr)` is LEAK (configuration GetWizardProgress, error_reporting GetErrorReport/GetCrashReport, auth Login/Refresh/ChangePassword via auth_service.go wraps, localization Import/Edit). Rule 2: text from a package-level sentinel or a `smartrules.ValidationError` is VALIDATION (playlist, service_handlers, conversion). Rule 3: `ErrLocalRootNotAllowed` text from `ResolveLocalRoot`/`CheckLocalRoot` is LEAK (it embeds `cannot resolve path: <OS error>`, local_policy.go:54). Rule 4: a binder (ShouldBindJSON) error on 400 is VALIDATION (mild: names Go struct fields). Rule 5: any other text on a 5xx or a variable status is LEAK.

## Class 1: SendErrorResponse family (the review's unlisted class)

`utils.SendErrorResponse(c, status, message, err)` at the review base (`utils/response.go:25-39`): if `err != nil`, `response.Details = err.Error()`, for every status. Bare-err call sites by file (review base):

| File | Sites |
|---|---|
| catalog-api/handlers/media_entity_handler.go | 35 |
| catalog-api/handlers/collection_handler.go | 15 |
| catalog-api/handlers/comic_pages_handler.go | 14 |
| catalog-api/handlers/media_handler.go | 10 |
| catalog-api/handlers/stats.go | 10 |
| catalog-api/handlers/recommendation_handler.go | 6 |
| catalog-api/handlers/search.go | 6 |
| catalog-api/handlers/browse.go | 5 |
| catalog-api/handlers/challenge.go | 5 |
| catalog-api/handlers/media_browse_handler.go | 2 |
| catalog-api/middleware/auth.go | 1 |
| **Total** | **109** in 11 files |

The review base has 167 SendErrorResponse call sites in non-test code: 109 pass a bare err variable (the table above; `nerr`/`rerr` at comic_pages_handler.go:582,592 are included), 1 passes `err.Error()` as the message (collection_handler.go:335), and 57 pass `nil` or a sentinel (`ErrRecommendationsNotImplemented`, recommendation_handler.go:115,152). The 57 carry no error text and are not rows.

HEAD: `SendErrorResponse` logs err (status, request id) and never sets `Details`; `ErrorResponse.Details` is no longer assigned in `utils/` (grep `Details =`: base 1 hit at response.go:31, HEAD 0), with `TestSendErrorResponse_DoesNotLeakInternalErrorText` and `..._HeadersDoNotLeakErrorText` in `utils/response_test.go`. All 109 rows are therefore CLOSED-BY-HELPER. Residual risk: the helper's `message` argument. At HEAD the `message` argument carries error-derived text at 2 SendErrorResponse sites: collection_handler.go:335 and :351, both `validationText(err)` (authored `ValidationError.Msg`, VALIDATION, listed in R4). The earlier sentence "0 TEXT rows" described only kinds `E:`/`F:`/`T:` and missed the helper kind `H:`; it is withdrawn. The 57 sites that pass `nil` or a sentinel include `ErrRecommendationsNotImplemented` (recommendation_handler.go:115,152): at the review base its own text "recommendation source not implemented" went into `details`; it is authored text, not internal detail (not counted as a row).

## Class 2: error text in struct/map literals and indirect flows

Found only by instrument 2 or by taint, not by the sink scan:

| Row | File:line (review base) | Flow | Class | HEAD |
|---|---|---|---|---|
| H1 | catalog-api/internal/metrics/health.go:157 | `Message: "Database ping failed: " + err.Error()` in a `ComponentHealth` returned by `checkDatabase`, serialised by the unauthenticated `/health/deep` route (registered at `main.go:1107` at base; the comment at `main.go:1052` says "/health and /health/deep stay public") | LEAK | CLOSED (fixed text "Database ping failed", error logged; CAT-PXF23) |
| H2 | catalog-api/internal/monitoring/monitoring.go:251 | `"error": err.Error()` in the map built by `HealthChecker.Check`, served by `GinHealthHandler` | LEAK, LATENT (package unwired) | STILL PRESENT, unchanged (see R1) |

H1 note: this is the only base instance of the "struct literal" route in `internal/metrics`. Instrument 2 lists exactly two `.Error()` calls in `internal/metrics` and `internal/monitoring` together at the review base (H1, H2), so the route is exhausted for those packages. user_handler.go:171,176 (`msg := err.Error()` echoed in `details`) are the "variable" route and appear in the per-site table as taint rows.

## Class 3: helpers that take an error (class-complete list)

Every non-test function in handlers, middleware, internal/handlers, internal/middleware, utils, internal/metrics and internal/monitoring with an error parameter was read (grep of `func ...(err error`). Result at HEAD:

| Helper | Does err text reach the client | Verdict |
|---|---|---|
| utils.SendErrorResponse | no (log only) | SAFE |
| handlers.respondInternal / respondBadBody / stdError (client_errors.go) | no (fixed message / status text) | SAFE |
| handlers.syncFail (client_errors.go:85) | fixed class text in `details` (syncDetails) | SAFE |
| handlers.validationText / sentinelText / paramRequiredText | authored text only (ValidationError.Msg, sentinel's own text, " parameter required") | VALIDATION |
| handlers.authClientMessage (auth_handler.go:39) | sentinel text only | VALIDATION |
| handlers.respondPDFError, writeLogServiceError, writeFixedLogError | fixed generic texts (already so at the review base) | SAFE |
| handlers.writeJobError (conversion_handler.go:337) | only `ErrConversionJobState` text | VALIDATION |
| handlers.rootURLMessage / localRootMessage / rootSpecMessage (scan_handler.go:553-578) | authored policy text; the caller's own path is echoed; OS error after "cannot resolve path" is dropped; URL refusal text is generic | VALIDATION (intentional) |
| handlers.AliasHandler.fail, middleware.useFallback, internal/handlers.smbErrorStatus | no (log only / fixed text) | SAFE |
| middleware.clientValidationDetails (input_validation.go:184) | only `validationError.public` | VALIDATION |
| smartrules.jsonErrText | authored ValidationError message built from the caller's own JSON decode error | VALIDATION |
| services notifyUser (conversion_service.go:958, sync_service.go:1084) | debug log only; `conversionNotifyHook` has no non-test setter | SAFE (not a client sink) |

At the review base the helpers that differed: `writeJobError` sent `err.Error()` for `ErrConversionJobState` (same authored text, VALIDATION), `authClientMessage` and the `client_errors.go` helpers did not exist (grep at the review base: 0 hits for `func authClientMessage` and no `client_errors.go`).

# Class 4: error text stored in state and served by another route

Found by instrument 3 (flow scan, below) and then traced by reading the writer, the model's JSON tag, the route registration and the gate. All routes are under the `/api/v1` group of `main.go` (authenticated unless a row says otherwise). Line numbers are HEAD (713a4f380) unless marked.

| Row | Field and writer | Served by (route, gate) | Scrub | Class |
|---|---|---|---|---|
| S1 | `ScanStatus.Reason`: `fail`/`cancel` in universal_scanner.go:505-522 store `scrubSecrets(err.Error(), secrets)` | `GET /scans`, `GET /scans/:job_id` (`scan_handler.go:539 "reason": s.Reason`; `RegisterScanRoutes`, storage_routes.go:25-26): any authenticated user. Also `publishScanEvent` (universal_scanner.go:161-194, payload `error`/`reason`) -> `EventBusBridge` (internal/handlers/eventbus_bridge.go:74,96,189-190; wired main.go:681,725) -> a `notification` message broadcast to all WebSocket clients (`BroadcastToClients`; the websocket route is authenticated, main.go:1151) | credentials and URL userinfo only (scan_redact.go); paths, host names and OS error text remain | LEAK (partly scrubbed), open |
| S2 | `GetStorageRootStatus` (scan_handler.go:332,339,345): `resp["error"] = services.ScrubErrorFor(err, root)`, HTTP 200 | `/storage-roots/:id/status`, no `RequireAdmin` (storage_routes.go:18): any authenticated user | `ScrubErrorFor` (same scrubber, plus the root's own secrets) | LEAK (partly scrubbed), open |
| S3 | `ListStoragePath` (scan_handler.go:378,384,401): same call | `/storage-roots/:id/list`, admin only (storage_routes.go:14) | same | LEAK to admins (partly scrubbed), accepted by role |
| S4 | `SyncSession.ErrorMessage`: `handleSyncError` (sync_service.go:1053-1066) stores raw `syncError.Error()`; chains such as "failed to get WebDAV client / scan local files / list remote files: %w" (models/user.go:1229, json `error_message,omitempty`) | `GET /sync/sessions`, `GET /sync/sessions/:id`, `GET /sync/history` (main.go:1657,1658,1715): the caller's own sessions, no admin gate | none. Measured: Go's net/http masks the password in a client error (`Get "http://usr:***@127.0.0.1:1/x": dial tcp ...`), so a password in the endpoint URL is masked by the standard client; other userinfo and transport routes are UNCONFIRMED. Host, path and OS text remain | LEAK, open |
| S5 | `ConversionJob.ErrorMessage` (models/file.go:158): conversion_service.go:884-885 stores `sanitizeConversionError(err.Error())` (reduces paths to base names) | `GET /conversion/jobs`, `GET /conversion/jobs/:id` (main.go:1410-1411): any authenticated user | paths reduced to base names; other text (driver, ffmpeg stderr) remains | LEAK (reduced), open |
| S6 | `WizardStepValidation.Errors[...]` (configuration_service.go:420,430,443): the three validators that can return an error return fixed authored texts ("path must be absolute", "invalid email format"), the database and network validators return nil; "%s is required" uses the field label | `POST /configuration/wizard/step/:step_id/validate` -> `ConfigurationServiceAdapter.ValidateWizardStep` (handlers/service_adapters.go:76) -> JSON (main.go:1495), gated by `system.configure` | none needed | VALIDATION |
| S7 | `ConfigurationImportResult.ValidationErrors`/`SkippedSettings` (internal/services/localization_service.go:849,857,877,886,895,904) | no route: `LocalizationHandlers` is constructed only in tests (`NewLocalizationHandlers` has 0 non-test callers; main.go registers no `/wizard/configuration` path) | n/a | LATENT |
| S8 | `BatchResult.Errors`, `SyncResult.Errors` (services/webdav_client.go:340,361,429,450,482,502) | no non-test caller of `UploadBatch`, `DownloadBatch`, `SyncDirectory` outside the file (grep: only webdav_client_test.go) | n/a | LATENT |
| S9 | `retry` result `Error` (internal/recovery/retry.go:401) | `internal/recovery` has no non-test importer (grep `internal/recovery"`: 0 hits; "digital.vasic.recovery" in internal/modules/registry.go is a name string) | n/a | LATENT |

| Row | Field and writer | Served by (route, gate) | Scrub | Class |
|---|---|---|---|---|
| S10 | `liveevidence.Record.ResolveError` (internal/liveevidence/liveevidence.go:158) | written to evidence files by Go tests; no non-test importer | n/a | SAFE (not a client sink) |
| S11 | `log_management_service.logError` (log_management_service.go:170,194,204): `message` is formatted from `%v` of errors | the function only calls the structured logger (lines 724-732) | n/a | SAFE |
| S12 | `notifyUser` (conversion_service.go, sync_service.go), `logSyncError`/`logSkippedFile` | log or debug output only | n/a | SAFE |

Instrument 3 reports 362 flow rows at HEAD (402 at the review base; `challenges/` is skipped, see R3). The field, key-value and call-argument rows of the files named in S1 to S12 were each traced above. The remaining rows (mostly constant or authored text assigned to fields, and calls into logging helpers) were NOT each traced one by one: UNCONFIRMED whether any of them reaches a client through a route this document did not look at (see Limits).

# HEAD residual (everything still in the class at 713a4f380)

| Id | Site | Class | Note |
|---|---|---|---|
| R1 | catalog-api/internal/monitoring/monitoring.go:251 `"error": err.Error()` (GinHealthHandler) | LATENT-OPEN | the package has no non-test importer; the leak becomes real the moment it is wired. Decide: sanitise or delete (delete needs operator confirmation, §11.4.122 / §11.4.124) |
| R2 | OCU-CUDA-Sidecar/internal/server/server.go:110,115,128,133,150,154,159,172,176,180,185,198,203,216,221 (15 sites, `http.Error(w, err.Error()...)` and `"bad request: "+err.Error()`) | LEAK | a separate Go module, same 15 sites at both bases, outside the catalog-api scan the old doc used. No in-module caller of `ListenAndServe` and no auth in `server.go`; whether the process is reachable by untrusted clients is UNCONFIRMED |
| R3 | catalog-api/handlers/challenge.go and the `challenges/` package: 155 `.Error()` calls (74 `CreateResult`, 20 `image_quality_final.go`, 16 `image_quality.go`, 13 `ch131_middleware_challenges.go`, ...) feed challenge result records that `ChallengeHandler.GetResults`, `RunChallenge` and `RunAll` return as JSON (`handlers/challenge.go:115,185`); `POST` run routes are admin-gated (`main.go:1670-1672`), the gating of `GET /api/v1/challenges/results` was not established | UNCONFIRMED | not classified per site; decide whether result text is operator-only |
| R4 | VALIDATION sentinel/authored sites: playlist_handler.go:95,99,188,273,320; collection_handler.go:335,351 (`validationText(err)` as the SendErrorResponse message); service_handlers.go:199,258,298 (`paramRequiredText(err)`); conversion_handler.go:344; scan_handler.go:178,186,194,448; middleware/input_validation.go:403 (`clientValidationDetails(err)`) | VALIDATION | authored text; scan rows echo the caller's own path; these rows are NOT closed, the text still reaches the client |
| R5 | `handlers/scan_handler.go` localRootMessage keeps `msg[:i] + ": cannot resolve path"` | VALIDATION | reads: the prefix of an authored policy sentence plus the path the caller sent |
| R6 | Class 4 rows S1, S2, S4, S5: `ScanStatus.Reason` (also on the WebSocket broadcast), `GetStorageRootStatus` `error`, `SyncSession.ErrorMessage`, `ConversionJob.ErrorMessage` | LEAK (partly scrubbed) | open; each is served to authenticated users. Decision owed per field: fixed class text for non-admin routes and the full text only for admins or the log. Not changed by this document |

Everything else the sink scan reports at HEAD is SAFE by the helper table (111 `SendErrorResponse` + 16 `respondInternal/respondBadBody` + 5 `respondPDFError` + 5 `writeLogServiceError` + 5 `writeJobError` + 2 `c.Error` bare-err rows; the `c.Error` rows are ErrorTypePrivate, which ErrorHandler answers with a fixed message at HEAD: `internal/middleware/middleware.go:97-100`).

## Census (instrument 2, all non-test code under catalog-api)

| Group | Review base | HEAD |
|---|---|---|
| production code | 175 | 97 |
| challenges/ (harness, see R3) | 155 | 155 |
| internal/tests (test helpers) | 3 | 3 |
| total | 333 | 255 |

HEAD production-code contexts (97): branching on text (`strings.Contains` 42, `strings.ToLower` 7, `strings.TrimPrefix` 2, `branch` 6) = 57, read at the branch: the compared text does not itself go to a response (stored-then-served flows are in Class 4); assignment 14, return 9, key-value 6, scrub/log helpers (`scrubSecrets` 5, `logSkippedFile` 2, `sanitizeConversionError` 1, `panic` 1, `paramRequiredRE.FindString` 1), `fmt.Sprintf` 1 (sync_service.go:1066, notifyUser, SAFE). `scrubSecrets` at universal_scanner.go:510,520 stores the scrubbed text in `ScanStatus.Reason`, which IS served (row S1); sync_service.go:1057-1058 `handleSyncError` stores the raw text in `SyncSession.ErrorMessage`, which IS served (row S4). The earlier sentences that none of these reach a response are withdrawn. The 14 assign + 9 return + 6 kv rows were read with the helper table above; the two that carry text to a body are the sentinel rows in R4 and the unwired R1.

## Review-base totals (instrument 1, 226 rows after removing `strings.*` callee noise) plus H1/H2

| Class | SendErrorResponse family | other sinks |
|---|---|---|
| LEAK | 109 | 49 |
| VALIDATION | 1 | 33 |
| LATENT | 0 | 15 |
| SAFE | 0 | 19 |
| total | 110 | 116 |

HEAD status of the 226 rows: CLOSED-BY-HELPER 109; CLOSED (err text no longer sent) 64 (49 LEAK + 15 LATENT); VALIDATION rows where `err.Error()` is no longer sent but authored text still is (NOT CLOSED) 32; unchanged (SAFE 19, VALIDATION 2) 21. This count covers the direct-sink rows only. It is NOT a claim that no leak remains: Class 4 rows S1 to S5 are still served at HEAD with credential-only or path-only scrubbing, and R1 to R3 are open.

## Per-site table (review base 9ed6577b, with HEAD status)

Row ids B1.. are sorted by file and line. `st` is the status argument when the call has one.

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B1 | catalog-api/handlers/asset_handler.go:91 | RequestAsset | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B2 | catalog-api/handlers/auth_handler.go:36 | LoginGin | c.JSON | http.StatusForbidden | `E:err.Error()` | VALIDATION | branch runs only when errors.Is(err, ErrAccountPending); Login returns the bare sentinel (auth_service.go:152 at base), so the body is its authored text "account pending approval" | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B3 | catalog-api/handlers/auth_handler.go:39 | LoginGin | c.JSON | http.StatusUnauthorized | `E:err.Error()` | LEAK | 401 branch: Login returns sentinels (invalid credentials, locked, disabled; authored) and also "failed to get user: %w" wrapping repository text (auth_service.go:132 at base); the repository text reaches the body | CLOSED (err text no longer sent) |
| B4 | catalog-api/handlers/auth_handler.go:59 | RefreshTokenGin | c.JSON | http.StatusUnauthorized | `E:err.Error()` | LEAK | (unresolved in review base) auth failure text; wording decides user-enumeration exposure | CLOSED (err text no longer sent) |
| B5 | catalog-api/handlers/auth_handler.go:76 | LogoutGin | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B6 | catalog-api/handlers/auth_handler.go:203 | RegisterGin | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B7 | catalog-api/handlers/auth_handler.go:321 | Login | http.Error | http.StatusUnauthorized | `E:err.Error()` | LATENT | (unresolved in review base) auth failure text; wording decides user-enumeration exposure. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...) | CLOSED (err text no longer sent) |
| B8 | catalog-api/handlers/auth_handler.go:346 | RefreshToken | http.Error | http.StatusUnauthorized | `E:err.Error()` | LATENT | (unresolved in review base) auth failure text; wording decides user-enumeration exposure. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...) | CLOSED (err text no longer sent) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B9 | catalog-api/handlers/auth_handler.go:368 | Logout | http.Error | http.StatusInternalServerError | `E:err.Error()` | LATENT | service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...) | CLOSED (err text no longer sent) |
| B10 | catalog-api/handlers/auth_handler.go:390 | LogoutAll | http.Error | http.StatusInternalServerError | `E:err.Error()` | LATENT | service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...) | CLOSED (err text no longer sent) |
| B11 | catalog-api/handlers/auth_handler.go:440 | ChangePassword | http.Error | http.StatusBadRequest | `E:err.Error()` | VALIDATION | deliberate validation text (may echo user input). The net/http `AuthHandler.ChangePassword` is also unrouted (main.go:1222 uses `mediaQueryHandler.ChangePassword`); kept VALIDATION (authored text), not recounted as LATENT | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B12 | catalog-api/handlers/auth_handler.go:446 | ChangePassword | http.Error | http.StatusBadRequest | `E:err.Error()` | LATENT | ChangePassword returns wrapped "failed to get user/ update password: %w" chains (auth_service.go:326,348,372) on 400. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...) | CLOSED (err text no longer sent) |
| B13 | catalog-api/handlers/auth_handler.go:468 | GetActiveSessions | http.Error | http.StatusInternalServerError | `E:err.Error()` | LATENT | service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...) | CLOSED (err text no longer sent) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B14 | catalog-api/handlers/auth_handler.go:507 | DeactivateSession | http.Error | http.StatusInternalServerError | `E:err.Error()` | LATENT | service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: net/http variant; main.go:1210-1220 registers only the Gin variants (LoginGin, RefreshTokenGin, LogoutGin, ...) | CLOSED (err text no longer sent) |
| B15 | catalog-api/handlers/browse.go:41 | GetStorageRoots | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B16 | catalog-api/handlers/browse.go:120 | BrowseDirectory | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B17 | catalog-api/handlers/browse.go:164 | GetFileInfo | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B18 | catalog-api/handlers/browse.go:217 | GetDirectorySizes | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B19 | catalog-api/handlers/browse.go:270 | GetDirectoryDuplicates | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B20 | catalog-api/handlers/challenge.go:103 | RunChallenge | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B21 | catalog-api/handlers/challenge.go:106 | RunChallenge | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B22 | catalog-api/handlers/challenge.go:166 | RunAll | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B23 | catalog-api/handlers/challenge.go:234 | RunByCategory | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B24 | catalog-api/handlers/challenge.go:237 | RunByCategory | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B25 | catalog-api/handlers/collection_handler.go:100 | loadAccessible | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B26 | catalog-api/handlers/collection_handler.go:105 | loadAccessible | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B27 | catalog-api/handlers/collection_handler.go:160 | ListCollections | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B28 | catalog-api/handlers/collection_handler.go:269 | GetCollection | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B29 | catalog-api/handlers/collection_handler.go:315 | ListCollectionItems | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B30 | catalog-api/handlers/collection_handler.go:335 | parseSmartRules | utils.SendErrorResponse | http.StatusBadRequest | `E:err.Error()` | VALIDATION | parseSmartRules: smartrules.ValidationError text passed as message (authored); non-validation errors take the 500 row at :337 | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B31 | catalog-api/handlers/collection_handler.go:337 | parseSmartRules | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B32 | catalog-api/handlers/collection_handler.go:353 | verifyRules | utils.SendErrorResponse | - | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B33 | catalog-api/handlers/collection_handler.go:374 | CreateCollection | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B34 | catalog-api/handlers/collection_handler.go:429 | CreateCollection | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B35 | catalog-api/handlers/collection_handler.go:452 | CreateCollection | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B36 | catalog-api/handlers/collection_handler.go:476 | UpdateCollection | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B37 | catalog-api/handlers/collection_handler.go:480 | UpdateCollection | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B38 | catalog-api/handlers/collection_handler.go:539 | UpdateCollection | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B39 | catalog-api/handlers/collection_handler.go:561 | DeleteCollection | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B40 | catalog-api/handlers/collection_handler.go:564 | DeleteCollection | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B41 | catalog-api/handlers/comic_pages_handler.go:106 | resolveArchive | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B42 | catalog-api/handlers/comic_pages_handler.go:113 | resolveArchive | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B43 | catalog-api/handlers/comic_pages_handler.go:116 | resolveArchive | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B44 | catalog-api/handlers/comic_pages_handler.go:434 | ListComicPages | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B45 | catalog-api/handlers/comic_pages_handler.go:467 | GetComicPage | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B46 | catalog-api/handlers/comic_pages_handler.go:478 | GetComicPage | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B47 | catalog-api/handlers/comic_pages_handler.go:492 | GetComicPage | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B48 | catalog-api/handlers/comic_pages_handler.go:515 | listCBRPages | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B49 | catalog-api/handlers/comic_pages_handler.go:522 | listCBRPages | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B50 | catalog-api/handlers/comic_pages_handler.go:552 | getCBRPage | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B51 | catalog-api/handlers/comic_pages_handler.go:559 | getCBRPage | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B52 | catalog-api/handlers/comic_pages_handler.go:570 | getCBRPage | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B53 | catalog-api/handlers/comic_pages_handler.go:582 | getCBRPage | utils.SendErrorResponse | http.StatusInternalServerError | `nerr` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B54 | catalog-api/handlers/comic_pages_handler.go:592 | getCBRPage | utils.SendErrorResponse | http.StatusInternalServerError | `rerr` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B55 | catalog-api/handlers/configuration_handler.go:57 | GetWizardStep | http.Error | http.StatusNotFound | `E:err.Error()` | VALIDATION | "wizard step not found: " authored, echoes the caller step id | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B56 | catalog-api/handlers/configuration_handler.go:77 | ValidateWizardStep | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B57 | catalog-api/handlers/configuration_handler.go:102 | SaveWizardProgress | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B58 | catalog-api/handlers/configuration_handler.go:121 | GetWizardProgress | http.Error | http.StatusNotFound | `E:err.Error()` | LEAK | GetWizardProgress returns repo text unchanged (driver / sql: no rows) on 404 | CLOSED (err text no longer sent) |
| B59 | catalog-api/handlers/configuration_handler.go:145 | CompleteWizard | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B60 | catalog-api/handlers/configuration_handler.go:173 | GetConfiguration | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B61 | catalog-api/handlers/configuration_handler.go:203 | TestConfiguration | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B62 | catalog-api/handlers/conversion_handler.go:73 | CreateJob | c.JSON | http.StatusBadRequest | `T:reason<-strings.TrimPrefix(strings.TrimPrefix(err.Error(), models.ErrInvalidConversionRe` | VALIDATION | INDIRECT: reason is the text after ErrInvalidConversionRequest (authored) | unchanged; class unchanged (VALIDATION) |
| B63 | catalog-api/handlers/conversion_handler.go:99 | GetJob | writeJobError | - | `err` | SAFE | helper answers fixed texts; only the ErrConversionJobState text is sent (row :344) | unchanged; class unchanged (SAFE) |
| B64 | catalog-api/handlers/conversion_handler.go:185 | CancelJob | writeJobError | - | `err` | SAFE | helper answers fixed texts; only the ErrConversionJobState text is sent (row :344) | unchanged; class unchanged (SAFE) |
| B65 | catalog-api/handlers/conversion_handler.go:246 | DeleteJob | writeJobError | - | `err` | SAFE | helper answers fixed texts; only the ErrConversionJobState text is sent (row :344) | unchanged; class unchanged (SAFE) |
| B66 | catalog-api/handlers/conversion_handler.go:282 | RetryJob | writeJobError | - | `err` | SAFE | helper answers fixed texts; only the ErrConversionJobState text is sent (row :344) | unchanged; class unchanged (SAFE) |
| B67 | catalog-api/handlers/conversion_handler.go:307 | DownloadJobFile | writeJobError | - | `err` | SAFE | helper answers fixed texts; only the ErrConversionJobState text is sent (row :344) | unchanged; class unchanged (SAFE) |
| B68 | catalog-api/handlers/conversion_handler.go:344 | writeJobError | c.JSON | http.StatusConflict | `E:err.Error()` | VALIDATION | ErrConversionJobState sentinel text, authored | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B69 | catalog-api/handlers/error_reporting_handler.go:85 | ReportError | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B70 | catalog-api/handlers/error_reporting_handler.go:123 | ReportCrash | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B71 | catalog-api/handlers/error_reporting_handler.go:152 | GetErrorReport | http.Error | http.StatusNotFound | `E:err.Error()` | LEAK | "failed to get error report: %w" wraps the repository error (driver text) | CLOSED (err text no longer sent) |
| B72 | catalog-api/handlers/error_reporting_handler.go:181 | GetCrashReport | http.Error | http.StatusNotFound | `E:err.Error()` | LEAK | "failed to get crash report: %w" wraps the repository error (driver text) | CLOSED (err text no longer sent) |
| B73 | catalog-api/handlers/error_reporting_handler.go:205 | ListErrorReports | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B74 | catalog-api/handlers/error_reporting_handler.go:232 | ListCrashReports | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B75 | catalog-api/handlers/error_reporting_handler.go:273 | UpdateErrorStatus | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B76 | catalog-api/handlers/error_reporting_handler.go:310 | UpdateCrashStatus | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B77 | catalog-api/handlers/error_reporting_handler.go:332 | GetErrorStatistics | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B78 | catalog-api/handlers/error_reporting_handler.go:355 | GetCrashStatistics | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B79 | catalog-api/handlers/error_reporting_handler.go:378 | GetSystemHealth | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B80 | catalog-api/handlers/error_reporting_handler.go:407 | UpdateConfiguration | http.Error | http.StatusInternalServerError | `E:err.Error()` | LATENT | service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: main.go registers none of UpdateConfiguration/ GetConfiguration/ CleanupOldReports/ ExportReports of ErrorReportingHandler (grep main.go: 0 hits) | CLOSED (err text no longer sent) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B81 | catalog-api/handlers/error_reporting_handler.go:432 | GetConfiguration | http.Error | http.StatusInternalServerError | `E:err.Error()` | LATENT | service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: main.go registers none of UpdateConfiguration/ GetConfiguration/ CleanupOldReports/ ExportReports of ErrorReportingHandler (grep main.go: 0 hits) | CLOSED (err text no longer sent) |
| B82 | catalog-api/handlers/error_reporting_handler.go:469 | CleanupOldReports | http.Error | http.StatusInternalServerError | `E:err.Error()` | LATENT | service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: main.go registers none of UpdateConfiguration/ GetConfiguration/ CleanupOldReports/ ExportReports of ErrorReportingHandler (grep main.go: 0 hits) | CLOSED (err text no longer sent) |
| B83 | catalog-api/handlers/error_reporting_handler.go:495 | ExportReports | http.Error | http.StatusInternalServerError | `E:err.Error()` | LATENT | service/repo/DB error text sent verbatim on 5xx or generic status. UNROUTED: main.go registers none of UpdateConfiguration/ GetConfiguration/ CleanupOldReports/ ExportReports of ErrorReportingHandler (grep main.go: 0 hits) | CLOSED (err text no longer sent) |
| B84 | catalog-api/handlers/log_management_handler.go:222 | GetLogEntries | writeLogServiceError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |
| B85 | catalog-api/handlers/log_management_handler.go:254 | CreateLogShare | writeLogServiceError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B86 | catalog-api/handlers/log_management_handler.go:322 | RevokeLogShare | writeLogServiceError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |
| B87 | catalog-api/handlers/log_management_handler.go:355 | ExportLogs | writeLogServiceError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |
| B88 | catalog-api/handlers/log_management_handler.go:475 | AnalyzeLogs | writeLogServiceError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |
| B89 | catalog-api/handlers/media_browse_handler.go:83 | SearchMedia | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B90 | catalog-api/handlers/media_browse_handler.go:106 | GetMediaStats | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B91 | catalog-api/handlers/media_browse_handler.go:127 | GetMediaStats | c.Error | - | `rowsErr` | SAFE | c.Error(rowsErr) is ErrorTypePrivate: ErrorHandler default branch answers a fixed message | unchanged; class unchanged (SAFE) |
| B92 | catalog-api/handlers/media_browse_handler.go:158 | GetMediaStats | c.Error | - | `rowsErr` | SAFE | same as :127 | unchanged; class unchanged (SAFE) |
| B93 | catalog-api/handlers/media_entity_handler.go:77 | SearchEntities | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B94 | catalog-api/handlers/media_entity_handler.go:85 | SearchEntities | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B95 | catalog-api/handlers/media_entity_handler.go:227 | ListEntities | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B96 | catalog-api/handlers/media_entity_handler.go:239 | ListEntities | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B97 | catalog-api/handlers/media_entity_handler.go:260 | GetEntity | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B98 | catalog-api/handlers/media_entity_handler.go:266 | GetEntity | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B99 | catalog-api/handlers/media_entity_handler.go:306 | GetEntityChildren | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B100 | catalog-api/handlers/media_entity_handler.go:315 | GetEntityChildren | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B101 | catalog-api/handlers/media_entity_handler.go:336 | GetEntityFiles | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B102 | catalog-api/handlers/media_entity_handler.go:342 | GetEntityFiles | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B103 | catalog-api/handlers/media_entity_handler.go:358 | GetEntityMetadata | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B104 | catalog-api/handlers/media_entity_handler.go:364 | GetEntityMetadata | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B105 | catalog-api/handlers/media_entity_handler.go:379 | GetEntityDuplicates | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B106 | catalog-api/handlers/media_entity_handler.go:385 | GetEntityDuplicates | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B107 | catalog-api/handlers/media_entity_handler.go:391 | GetEntityDuplicates | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B108 | catalog-api/handlers/media_entity_handler.go:415 | GetEntityTypes | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B109 | catalog-api/handlers/media_entity_handler.go:457 | BrowseByType | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B110 | catalog-api/handlers/media_entity_handler.go:463 | BrowseByType | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B111 | catalog-api/handlers/media_entity_handler.go:496 | RefreshEntityMetadata | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B112 | catalog-api/handlers/media_entity_handler.go:1158 | entityMustExist | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B113 | catalog-api/handlers/media_entity_handler.go:1160 | entityMustExist | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B114 | catalog-api/handlers/media_entity_handler.go:1173 | UpdateUserMetadata | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B115 | catalog-api/handlers/media_entity_handler.go:1186 | UpdateUserMetadata | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B116 | catalog-api/handlers/media_entity_handler.go:1217 | UpdateUserMetadata | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B117 | catalog-api/handlers/media_entity_handler.go:1246 | StreamEntity | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B118 | catalog-api/handlers/media_entity_handler.go:1260 | StreamEntity | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B119 | catalog-api/handlers/media_entity_handler.go:1263 | StreamEntity | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B120 | catalog-api/handlers/media_entity_handler.go:1283 | DownloadEntity | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B121 | catalog-api/handlers/media_entity_handler.go:1291 | DownloadEntity | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B122 | catalog-api/handlers/media_entity_handler.go:1320 | DownloadEntity | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B123 | catalog-api/handlers/media_entity_handler.go:1323 | DownloadEntity | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B124 | catalog-api/handlers/media_entity_handler.go:1343 | GetInstallInfo | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B125 | catalog-api/handlers/media_entity_handler.go:1349 | GetInstallInfo | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B126 | catalog-api/handlers/media_entity_handler.go:1369 | GetInstallInfo | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B127 | catalog-api/handlers/media_entity_handler.go:1402 | ListDuplicateGroups | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B128 | catalog-api/handlers/media_handler.go:46 | GetMediaByID | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B129 | catalog-api/handlers/media_handler.go:87 | GetMediaByID | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B130 | catalog-api/handlers/media_handler.go:89 | GetMediaByID | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B131 | catalog-api/handlers/media_handler.go:148 | UpdateWatchProgress | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B132 | catalog-api/handlers/media_handler.go:155 | UpdateWatchProgress | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B133 | catalog-api/handlers/media_handler.go:205 | UpdateWatchProgress | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B134 | catalog-api/handlers/media_handler.go:238 | UpdateFavoriteStatus | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B135 | catalog-api/handlers/media_handler.go:245 | UpdateFavoriteStatus | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B136 | catalog-api/handlers/media_handler.go:267 | UpdateFavoriteStatus | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B137 | catalog-api/handlers/media_handler.go:274 | UpdateFavoriteStatus | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B138 | catalog-api/handlers/pdf_pages_handler.go:101 | resolvePDF | respondPDFError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B139 | catalog-api/handlers/pdf_pages_handler.go:282 | ListPdfPages | respondPDFError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |
| B140 | catalog-api/handlers/pdf_pages_handler.go:289 | ListPdfPages | respondPDFError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |
| B141 | catalog-api/handlers/pdf_pages_handler.go:322 | GetPdfPage | respondPDFError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |
| B142 | catalog-api/handlers/pdf_pages_handler.go:329 | GetPdfPage | respondPDFError | - | `err` | SAFE | helper maps errors to a fixed, generic body at base (pdf_pages_handler.go:230-258, log_management_handler.go:35-49); bare err argument is not serialised | unchanged; class unchanged (SAFE) |
| B143 | catalog-api/handlers/playback_handler.go:79 | StartSession | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B144 | catalog-api/handlers/playback_handler.go:96 | StartSession | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B145 | catalog-api/handlers/playback_handler.go:106 | ProgressSession | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B146 | catalog-api/handlers/playback_handler.go:118 | ProgressSession | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B147 | catalog-api/handlers/playback_handler.go:128 | EndSession | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B148 | catalog-api/handlers/playback_handler.go:143 | EndSession | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B149 | catalog-api/handlers/playback_handler.go:170 | GetProgressForEntity | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B150 | catalog-api/handlers/playback_handler.go:193 | ListHistoryForEntity | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B151 | catalog-api/handlers/playlist_handler.go:95 | CreatePlaylist | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | smartrules.IsValidation branch: authored ValidationError text (may echo caller JSON) | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B152 | catalog-api/handlers/playlist_handler.go:99 | CreatePlaylist | c.JSON | http.StatusConflict | `E:err.Error()` | VALIDATION | ErrSmartLimit sentinel with authored count text | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B153 | catalog-api/handlers/playlist_handler.go:188 | UpdatePlaylist | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | smartrules.IsValidation branch: authored text | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B154 | catalog-api/handlers/playlist_handler.go:273 | AddItem | c.JSON | http.StatusConflict | `E:err.Error()` | VALIDATION | ErrSmartPlaylistItems sentinel, authored | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B155 | catalog-api/handlers/playlist_handler.go:320 | RemoveItem | c.JSON | http.StatusConflict | `E:err.Error()` | VALIDATION | ErrSmartPlaylistItems sentinel, authored | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B156 | catalog-api/handlers/recommendation_handler.go:54 | GetSimilarItems | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B157 | catalog-api/handlers/recommendation_handler.go:76 | GetSimilarItems | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B158 | catalog-api/handlers/recommendation_handler.go:93 | GetSimilarItems | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B159 | catalog-api/handlers/recommendation_handler.go:115 | GetTrendingItems | utils.SendErrorResponse | http.StatusNotImplemented | `ErrRecommendationsNotImplemented` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B160 | catalog-api/handlers/recommendation_handler.go:135 | GetPersonalizedRecommendations | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B161 | catalog-api/handlers/recommendation_handler.go:152 | GetPersonalizedRecommendations | utils.SendErrorResponse | http.StatusNotImplemented | `ErrRecommendationsNotImplemented` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B162 | catalog-api/handlers/scan_handler.go:153 | CreateStorageRoot | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B163 | catalog-api/handlers/scan_handler.go:177 | CreateStorageRoot | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | deliberate validation text (may echo user input) | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B164 | catalog-api/handlers/scan_handler.go:185 | CreateStorageRoot | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | deliberate validation text (may echo user input) | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B165 | catalog-api/handlers/scan_handler.go:193 | CreateStorageRoot | c.JSON | http.StatusForbidden | `E:err.Error()` | LEAK | ErrLocalRootNotAllowed text includes "cannot resolve path: " and server-side policy detail (local_policy.go:54,48,63) | CLOSED (err text no longer sent) |
| B166 | catalog-api/handlers/scan_handler.go:227 | CreateStorageRoot | c.JSON | http.StatusInternalServerError | `F:fmt.Sprintf("failed to create storage root: %v", insertErr)` | LEAK | %v of DB/scanner error wrapped into body (SQL/driver detail) | CLOSED (err text no longer sent) |
| B167 | catalog-api/handlers/scan_handler.go:247 | GetStorageRoots | c.JSON | http.StatusInternalServerError | `F:fmt.Sprintf("failed to query storage roots: %v", err)` | LEAK | %v of DB/scanner error wrapped into body (SQL/driver detail) | CLOSED (err text no longer sent) |
| B168 | catalog-api/handlers/scan_handler.go:427 | QueueScan | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B169 | catalog-api/handlers/scan_handler.go:441 | QueueScan | c.JSON | http.StatusNotFound | `F:fmt.Sprintf("storage root not found: %v", err)` | LEAK | %v of DB/scanner error wrapped into body (SQL/driver detail) | CLOSED (err text no longer sent) |
| B170 | catalog-api/handlers/scan_handler.go:447 | QueueScan | c.JSON | http.StatusForbidden | `E:err.Error()` | LEAK | same producer as :193 (CheckLocalRoot) | CLOSED (err text no longer sent) |
| B171 | catalog-api/handlers/scan_handler.go:463 | QueueScan | c.JSON | http.StatusServiceUnavailable | `F:fmt.Sprintf("failed to queue scan: %v", err)` | LEAK | %v of DB/scanner error wrapped into body (SQL/driver detail) | CLOSED (err text no longer sent) |
| B172 | catalog-api/handlers/search.go:100 | SearchFiles | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B173 | catalog-api/handlers/search.go:108 | SearchFiles | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B174 | catalog-api/handlers/search.go:155 | SearchFiles | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B175 | catalog-api/handlers/search.go:258 | SearchDuplicates | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B176 | catalog-api/handlers/search.go:287 | AdvancedSearch | utils.SendErrorResponse | http.StatusBadRequest | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B177 | catalog-api/handlers/search.go:328 | AdvancedSearch | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B178 | catalog-api/handlers/service_handlers.go:199 | CreateReport | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | ErrUnsupportedReportType sentinel, authored | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B179 | catalog-api/handlers/service_handlers.go:258 | GetUsageReport | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | body only reached when text matches "parameter required" (deliberate); other errors get generic 500 | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B180 | catalog-api/handlers/service_handlers.go:298 | GetPerformanceReport | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | body only reached when text matches "parameter required" (deliberate); other errors get generic 500 | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B181 | catalog-api/handlers/stats.go:42 | GetOverallStats | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B182 | catalog-api/handlers/stats.go:82 | GetSmbRootStats | utils.SendErrorResponse | http.StatusNotFound | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B183 | catalog-api/handlers/stats.go:85 | GetSmbRootStats | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B184 | catalog-api/handlers/stats.go:119 | GetFileTypeStats | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B185 | catalog-api/handlers/stats.go:146 | GetSizeDistribution | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B186 | catalog-api/handlers/stats.go:173 | GetDuplicateStats | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B187 | catalog-api/handlers/stats.go:214 | GetTopDuplicateGroups | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B188 | catalog-api/handlers/stats.go:248 | GetAccessPatterns | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B189 | catalog-api/handlers/stats.go:282 | GetGrowthTrends | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |
| B190 | catalog-api/handlers/stats.go:321 | GetScanHistory | utils.SendErrorResponse | http.StatusInternalServerError | `err` | LEAK | err.Error() copied into the response "details" field by utils.SendErrorResponse (response.go:31), any status | CLOSED-BY-HELPER (details no longer set; err logged) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B191 | catalog-api/handlers/sync_handler.go:40 | CreateEndpoint | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B192 | catalog-api/handlers/sync_handler.go:52 | CreateEndpoint | c.JSON | - | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B193 | catalog-api/handlers/sync_handler.go:69 | GetUserEndpoints | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B194 | catalog-api/handlers/sync_handler.go:103 | GetEndpoint | c.JSON | - | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B195 | catalog-api/handlers/sync_handler.go:126 | UpdateEndpoint | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B196 | catalog-api/handlers/sync_handler.go:142 | UpdateEndpoint | c.JSON | - | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B197 | catalog-api/handlers/sync_handler.go:171 | DeleteEndpoint | c.JSON | - | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B198 | catalog-api/handlers/sync_handler.go:202 | StartSync | c.JSON | - | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B199 | catalog-api/handlers/sync_handler.go:234 | GetUserSessions | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B200 | catalog-api/handlers/sync_handler.go:268 | GetSession | c.JSON | - | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B201 | catalog-api/handlers/sync_handler.go:288 | ScheduleSync | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B202 | catalog-api/handlers/sync_handler.go:304 | ScheduleSync | c.JSON | - | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B203 | catalog-api/handlers/sync_handler.go:341 | GetSyncStatistics | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B204 | catalog-api/handlers/sync_handler.go:371 | CleanupOldSessions | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B205 | catalog-api/handlers/user_handler.go:121 | CreateUser | http.Error | http.StatusBadRequest | `E:err.Error()` | VALIDATION | deliberate validation text (may echo user input) | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B206 | catalog-api/handlers/user_handler.go:171 | CreateUser | http.Error | http.StatusConflict | `T:msg<-err.Error()` | LEAK | INDIRECT: msg := err.Error() is sent as "details" on 409 and 500 (user_handler.go:172,177 at base): UNIQUE/PG duplicate-key text and any repo error | CLOSED (details now fixed text; msg only branches, user_handler.go:168-179) |
| B207 | catalog-api/handlers/user_handler.go:176 | CreateUser | http.Error | http.StatusInternalServerError | `T:msg<-err.Error()` | LEAK | INDIRECT: same msg as :171 sent as details on 500 | CLOSED (details now fixed text; msg only branches, user_handler.go:168-179) |
| B208 | catalog-api/handlers/user_handler.go:624 | ResetPassword | http.Error | http.StatusBadRequest | `E:err.Error()` | VALIDATION | deliberate validation text (may echo user input) | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B209 | catalog-api/handlers/user_handler.go:630 | ResetPassword | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B210 | catalog-api/handlers/user_handler.go:694 | LockAccount | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B211 | catalog-api/handlers/user_handler.go:738 | UnlockAccount | http.Error | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B212 | catalog-api/internal/handlers/copy.go:249 | CopyToStorage | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B213 | catalog-api/internal/handlers/localization_handlers.go:301 | ImportConfiguration | h.sendError | http.StatusBadRequest | `F:fmt.Sprintf("Failed to import configuration: %v", err)` | LATENT | mixed: "failed to parse configuration JSON: %w" plus service/DB wrapping, sent via %v. UNROUTED: `LocalizationHandlers` is constructed only in `internal/handlers/localization_handlers_test.go`; no non-test registration (grep NewLocalizationHandlers / wizard/configuration in main.go: 0 hits) | CLOSED (err text no longer sent) |
| B214 | catalog-api/internal/handlers/localization_handlers.go:365 | EditConfiguration | h.sendError | http.StatusBadRequest | `F:fmt.Sprintf("Failed to edit configuration: %v", err)` | LATENT | mixed: service error wrapped via %v into a 400 body. UNROUTED: `LocalizationHandlers` is constructed only in `internal/handlers/localization_handlers_test.go`; no non-test registration (grep NewLocalizationHandlers / wizard/configuration in main.go: 0 hits) | CLOSED (err text no longer sent) |
| B215 | catalog-api/internal/handlers/smb_discovery.go:86 | DiscoverShares | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B216 | catalog-api/internal/handlers/smb_discovery.go:124 | TestConnection | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B217 | catalog-api/internal/handlers/smb_discovery.go:176 | BrowseShare | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | binding/JSON-decode error; names Go struct fields/types, echoes malformed input | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B218 | catalog-api/internal/middleware/middleware.go:81 | ErrorHandler | c.JSON | http.StatusBadRequest | `E:err.Error()` | LATENT | ErrorTypeBind branch; ErrorHandler is wired (main.go:1038) but no non-test code produces a gin bind error in c.Errors (no Bind/BindJSON/MustBindWith; ShouldBind* does not append) | CLOSED (err text no longer sent) |

| # | File:line | Function | Sink | st | Expression | Class | Reason | HEAD |
|---|---|---|---|---|---|---|---|---|
| B219 | catalog-api/internal/middleware/middleware.go:86 | ErrorHandler | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LATENT | ErrorTypePublic branch; no non-test code sets ErrorTypePublic (grep: 0 hits outside middleware.go) | CLOSED (err text no longer sent) |
| B220 | catalog-api/internal/smartrules/smartrules.go:207 | parseList | jsonErrText | - | `err` | VALIDATION | jsonErrText builds the authored ValidationError message from the caller own JSON decode error | unchanged; class unchanged (VALIDATION) |
| B221 | catalog-api/main.go:1306 | main | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B222 | catalog-api/main.go:1337 | main | c.JSON | http.StatusInternalServerError | `E:err.Error()` | LEAK | service/repo/DB error text sent verbatim on 5xx or generic status | CLOSED (err text no longer sent) |
| B223 | catalog-api/middleware/auth.go:167 | RequireAuth | utils.SendErrorResponse | http.StatusUnauthorized | `err` | LEAK | JWT parse text (token, signing-method and claims detail) sent as details on 401 | CLOSED-BY-HELPER (details no longer set; err logged) |
| B224 | catalog-api/middleware/input_validation.go:360 | InputValidation | c.JSON | http.StatusBadRequest | `E:err.Error()` | VALIDATION | deliberate validation text (may echo user input) | err.Error() no longer sent; NOT CLOSED: HEAD still sends authored text through a helper, or a fixed text (see R4) |
| B225 | catalog-api/services/conversion_service.go:903 | handleConversionError | s.notifyUser | - | `F:fmt.Sprintf("Conversion failed: %s", errorMsg)` | SAFE | notifyUser writes a debug log line and calls conversionNotifyHook (test-only setter; no production writer found); not a client sink | unchanged; class unchanged (SAFE) |
| B226 | catalog-api/services/sync_service.go:1066 | handleSyncError | s.notifyUser | - | `E:syncError.Error()` | SAFE | notifyUser writes a debug log line only (sync_service.go:1084-1093); not a client sink | unchanged; class unchanged (SAFE) |

| H1 | catalog-api/internal/metrics/health.go:157 | checkDatabase | struct literal | - | `"Database ping failed: " + err.Error()` | LEAK | unauthenticated /health/deep body | CLOSED (fixed text, error logged) | | H2 | catalog-api/internal/monitoring/monitoring.go:251 | Check | map literal | - | `err.Error()` | LEAK (latent) | package unwired | STILL PRESENT (R1) |

# Limits of the instruments (measured, UNCONFIRMED where stated)

- Type information is heuristic: any zero-argument `.Error()` matches, including non-error types (`writer.Error()` of encoding/csv appears in the census; read, not a response); err-like identifiers are found by declared type `error` or a nil comparison, not by the type checker, so an `error` value obtained from a multi-value call and never compared with nil, a field of type `error`, or an interface method result is missed (round-2 blind spot, partly closed).

- Sink detection is by callee name; a response writer with a name outside the list is missed by instrument 1 and is caught only if instrument 2 sees an `.Error()` inside it. Error text built without `.Error()`, `%v` of an err-named identifier, or an err-named identifier argument (for instance a variable named `cause` or `e` passed to `fmt.Sprintf`) is not found by either instrument: UNCONFIRMED whether any exists.
- Blind spots found by the round-2 review and closed only as far as stated: helper-in-sink (`H:`), non-err-named variable in `%v` (`F:`). Still NOT covered: error text round-tripped through the database or a file and served by a different route than the writer's, event-bus consumers other than `EventBusBridge`, text built by `errors.Unwrap`/`errors.Join` chains and then formatted with `%s` of a non-error value, `fmt.Errorf` results returned to a handler that sends them (covered only when the handler's sink argument shows an err-like identifier), and routes registered through dynamic mux patterns. Class 4 is a trace of the fields instrument 3 surfaced, not a proof that no other field is served.
- Taint is flow-insensitive within a function (one false positive found and read, user_handler.go:122 at HEAD). Taint through struct fields, other functions or channels is not followed; the census catches the `.Error()` call itself but not where the stored text is later sent.
- BARE reports any err-named identifier argument to a sink-named callee, including calls whose helper never serialises it (that is why 111 SendErrorResponse rows are listed as CLOSED-BY-HELPER rather than as leaks).
- Producers: service and repository error strings were read only where a row says so (auth_service.go, configuration_service.go, error_reporting_service.go, local_policy.go, localization_service.go, playlist_service.go). Every other LEAK means "text can include internal detail", not a demonstrated leak.
- Not covered: Go modules other than catalog-api and OCU-CUDA-Sidecar (examples/, docs/qa/lib/ctrctl, catalog-api/tests/automation, tools/ fixtures were not scanned), JavaScript/TypeScript backends, generated code, and HTTP responses produced by third-party middleware. WebSocket (`handlers/websocket_handler.go`) and SSE (`log_management_handler.go`) code was grepped: error values there go to `h.logger` only (UNCONFIRMED beyond that grep).
- HEAD status was produced by matching each review-base row to the HEAD scan by (file, function, sink, kind, expression) and then reading the helper and the changed line for the rows that moved; line numbers shift, so the match ignores them.